Shifting Macroeconomic Tides: Investigating the Impact of Interest Rate Uncertainty on the Canadian Economy

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Abstract

This paper investigates the macroeconomic impact of interest rate uncertainty on the Canadian economy using a Vector Autoregression with Exogenous Variables (VAR-X) framework. Constructing a market-based proxy from the quarterly volatility of three-month treasury bill returns over the 1961–2023 period, I find that heightened uncertainty exerts a consistent, statistically significant contractionary effect on real GDP and fixed capital formation. Furthermore, lagged specifications outperform contemporaneous ones for investment when broader uncertainty controls are included, lending empirical support to the "wait-and-see" mechanism proposed by real options theory. In contrast, the response of nominal variables remains ambiguous. Interest rates exhibit inconsistent coefficient signs across specifications, likely reflecting bidirectional causality between uncertainty and policy responses, while inflation shows no statistically significant effects, suggesting that supply-side and demand-side channels may be offsetting.

JEL Classifications: E44; E50
Keywords: Interest Rates, Macroeconomic Fluctuations, Uncertainty, Canada

1 Introduction

Academic interest in the economic dynamics of uncertainty has surged in recent decades. It is widely established that uncertainty shocks have a significant negative impact on the real economy (Leduc and Liu 2016). However, while this theoretical framework is well-established, empirical research on monetary policy uncertainty remains predominantly US-focused, leaving the Canadian context largely unexplored. Moreover, even the limited existing Canadian research faces methodological constraints, with the sole study (Istrefi and Mouabbi 2016) relying exclusively on subjective survey-based proxies for interest rate uncertainty.

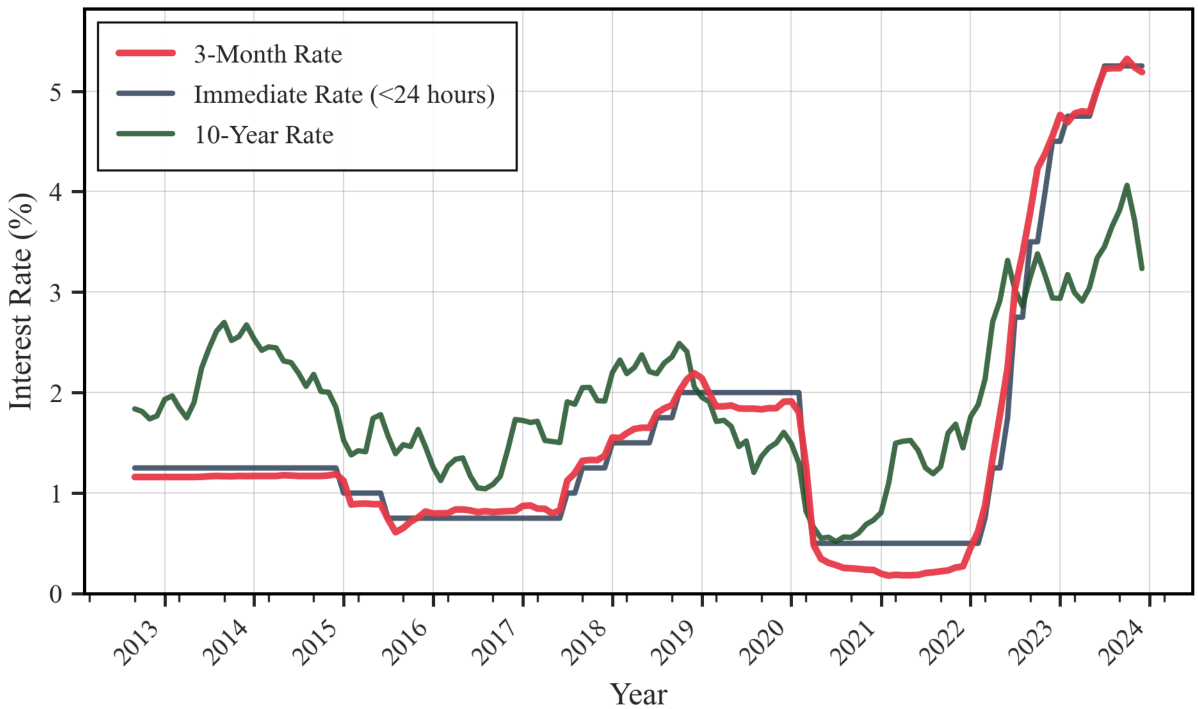

Figure 1: Canadian Interest Rates From 2012 to 2023

This knowledge gap has become particularly salient following recent macroeconomic disruptions. Prior to the COVID-19 pandemic, Canada experienced a prolonged period of low interest rates, with short-term and long-term rates maintained within the 1-3 percent range during the 2012-2020 period (see Figure 1). However, the overlapping supply and demand shocks triggered by the COVID-19 pandemic, amplified by the Bank of Canada's near-zero interest rate policy, contributed to persistent inflation. In March 2022, the Bank of Canada implemented substantial rate hikes to rein in these pressures. This sequence of rapid reductions followed by unprecedented increases ushered in an era of heightened interest rate uncertainty. Businesses and households, having adapted to cheap credit during the prolonged low-rate period, now face the challenge of recalibrating their expectations and strategies in response to evolving interest rate dynamics. Against this backdrop, this study uses quarterly volatility in daily returns for three-month Canadian treasury bills to evaluate interest rate uncertainty's impact on gross domestic product, fixed capital formation, inflation, and short-term interest rates.

2 Related Literature

2.1 Impact of Uncertainty

While researchers employ different estimation methods, most find similar results regarding interest rate uncertainty's impact on the economy. Focusing on the United States, Qadan et al. (2023) find that interest rate uncertainty is linked to higher unemployment and contractions in industrial output. These results are similar to those of Bretscher et al. (2018) who find that a one standard deviation change in their proxy of interest rate uncertainty for the United States results in a $52 billion drop in aggregate investment. Istrefi and Mouabbi (2016) also observe that interest rate uncertainty shocks are recessionary, increasing unemployment while reducing production and prices.

2.2 Uncertainty Indicators

Another area of interest pertains to identifying the most suitable proxy for measuring interest rate uncertainty. One approach uses keyword frequency counts, with Husted et al. (2019) gauging monetary policy uncertainty by tracking newspaper article and keyword frequency related to interest rates. Survey-based approaches have also become increasingly preferred, with disagreement among forecasters serving as a frequently employed proxy (Rich and Tracy 2010; Dovern et al. 2012). However, the validity of such proxies is contested, as disagreement may reflect differences in opinion rather than uncertainty (Diether et al. 2002), and proxy reliability depends on forecasting environment stability, which can introduce bias (Lahiri and Sheng 2010). To address these challenges, Istrefi and Mouabbi (2016), building on D'Amico and Orphanides (2014), introduce a subjective ex-ante interest rate uncertainty proxy accounting for both disagreement among forecasters and perceived variability of future aggregate shocks. Regardless of these refinements, a binding constraint remains: the reliance on a restricted pool of market participants may inadequately represent the broader economy (Chang and Feunou 2013). An alternative approach employs option price volatility to evaluate interest rate uncertainty. Most papers that adopt this approach use implied volatility from interest rate futures options (Neely 2005; Bauer et al. 2021), while some use realized volatility computed from intraday price movements (Kaminska and Roberts-Sklar 2018). Chang and Feunou (2013) provide the only Canadian-focused measure, formulating volatility based on futures contracts associated with the three-month bankers' acceptance rate (BAX) and related options (OBX).

2.3 Uncertainty Transmission Channels

Previous research indicates that monetary policy uncertainty typically reduces investment, employment, and real GDP (Istrefi and Mouabbi 2016; Jiang et al. 2022), though the specific transmission channels remain debated. Real option theory suggests that economic agents factor macroeconomic uncertainties into investment decisions (Davig and Hakkio, 2010), with uncertain periods making private sector entities increasingly risk-averse and prompting investment deferrals (Panousi and Papanikolaou, 2012). More uncertain markets increase call option values, leading firms to adopt a "wait-and-watch" approach (Bernanke, 1983), which then further intensifies uncertainty's effect on real activity (Bloom, 2009). Uncertainty also operates through financial friction channels, where increased uncertainty raises credit spreads (Higgins, 2023), reducing lending and investment, while elevated risk levels result in larger bankruptcy losses that force household consumption cuts (Gilchrist et al. 2014). Moreover, banks deleverage when uncertainty rises, further disrupting financial markets and curtailing credit availability to firms (Istiak and Serletis, 2020). Researchers have also identified precautionary savings as a transmission channel, whereby individuals facing uncertainty adopt conservative consumption approaches, increasing savings (Lugilde et al. 2017), reducing current consumption, and ultimately contracting real GDP (Luo et al. 2022).

3 Conceptual Framework

To investigate interest rate uncertainty's impact on macroeconomic variables, I employ a Vector Autoregressive with exogenous variables (VAR-X) model, first introduced by Pesaran, Shin, and Smith (2001). I specify Real Gross Domestic Product (GDP), Business Gross Fixed Capital Formation (GFCF), Inflation (π), and 3-month Interest Rates (IR) as endogenous variables. For my exogenous uncertainty measure, motivated by Arnold and Vrugt (2010), who document that treasury bond volatility is primarily determined by monetary policy uncertainty, I develop a market-based proxy using the volatility of daily returns for 3-month treasury bills (TBV). [1] This approach addresses the limitations of survey-based and keyword count measures, which rely on restricted participant samples, and avoids the extensive derivatives market data required for implied volatility measures.

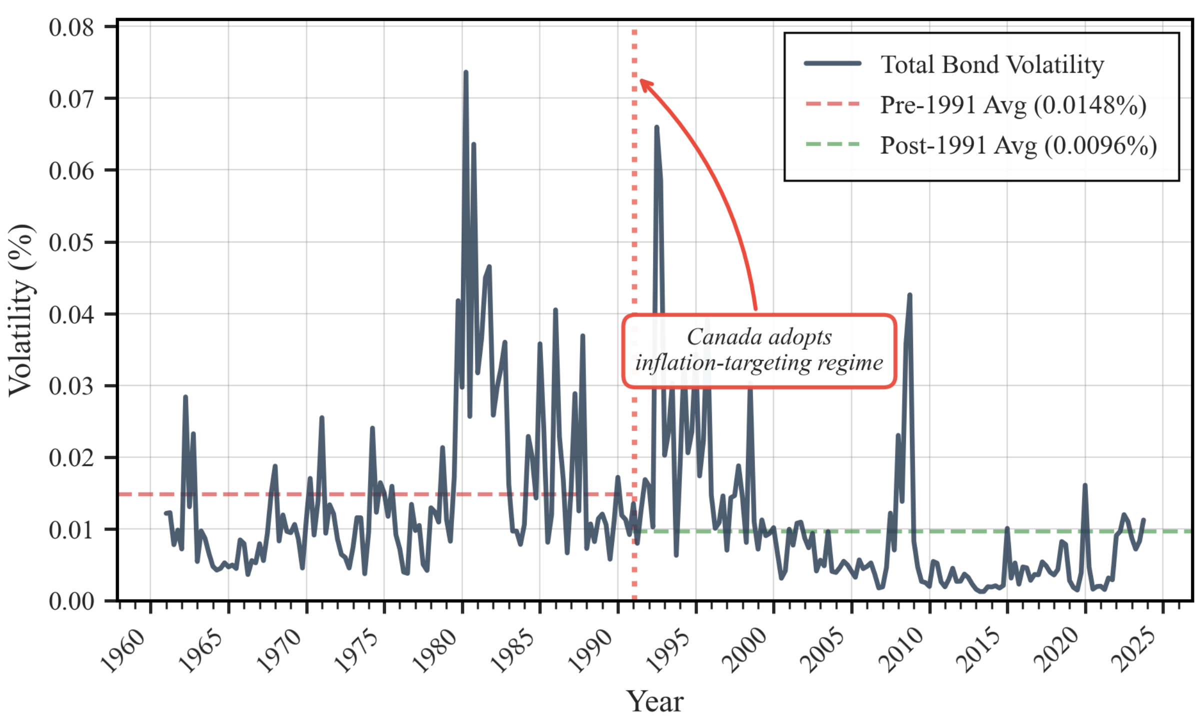

Figure 2: Quarterly Total Bond Volatility (1961–2023)

[1]See Appendix A for detailed calculation methodology.

As depicted in Figure 2, TBV closely mirrors significant disruptions in the Canadian economy. The measure exhibits spikes during the Volcker disinflation era (Q4 1979, Q2 1980), the early 1990s recession (Q1-Q2 1991), and the 2008 global financial crisis (Q3-Q4 2008), confirming its efficacy in capturing heightened uncertainty periods. Beyond discrete crisis episodes, TBV also responds to structural policy changes, with average volatility declining by 36% following Canada's adoption of inflation targeting in February 1991, suggesting the measure captures reduced monetary policy uncertainty under a more credible monetary policy regime. In addition to aligning with historical events, TBV also shows strong consistency with established alternative measures. As shown in Figure 3, TBV correlates closely with Istrefi and Mouabbi's (2016) survey-based proxy, which tracks disagreements among forecasters.

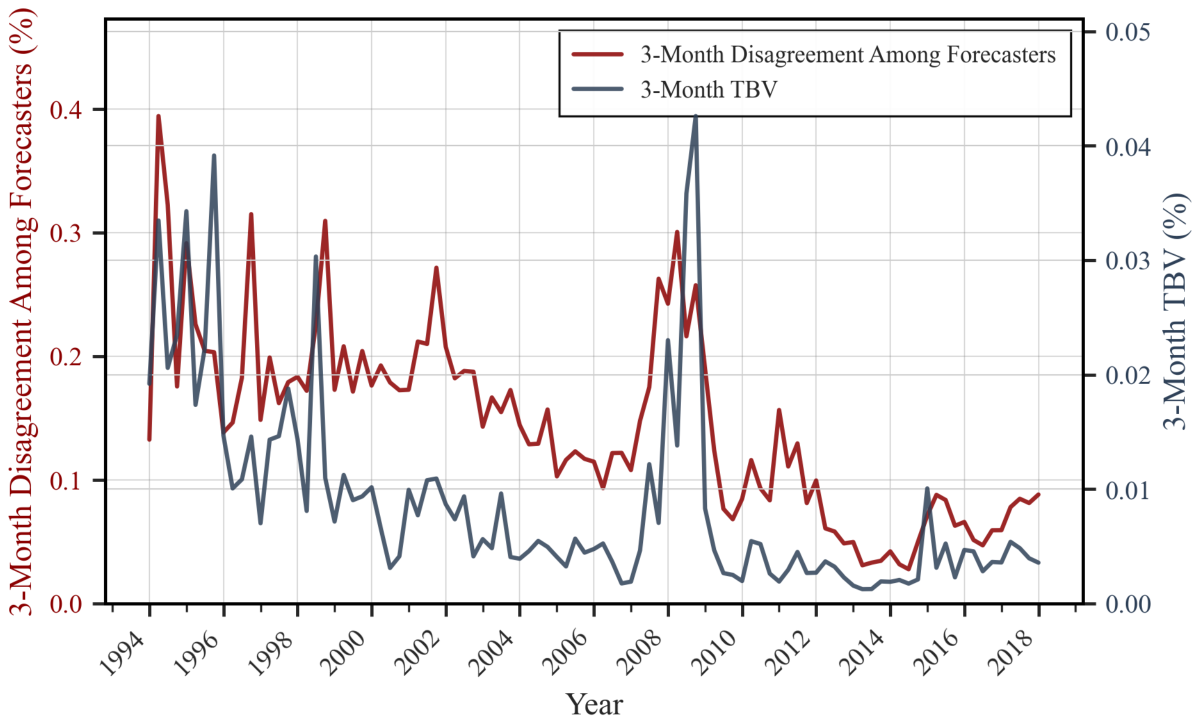

Figure 3: Comparison: TBV vs. Disagreement Among Forecasters (1994–2018). The two series exhibit a Pearson correlation coefficient of $\rho = 0.68$ (p-value < 0.01).

In addition to TBV, I also incorporate Economic Policy Uncertainty (EPU) and Oil Price Uncertainty (OPU) as exogenous variables to control for broader economic uncertainty and Canada's oil export dependence, respectively. I estimate six VAR-X specifications following the general form:

$$\mathbf{Y}_t = \mathbf{A}_0 + \mathbf{A}_1 \mathbf{Y}_{t-1} + \mathbf{\Phi} \mathbf{X}_t + \epsilon_t \tag{1}$$

where $\mathbf{Y}_t = [\Delta GDP_t, \Delta GFCF_t, \Delta IR_t, \pi_t]'$ is the vector of endogenous variables[2]. The exogenous vector $\mathbf{X}_t$ and sample period vary by specification (Table 1). Models 1 and 2 are estimated over both the full sample (1961–2023) and a restricted period (1985–2023) to enable comparison with Models 3–4, which require EPU and OPU data available only from 1985.

[2] ΔGDP_t and $\Delta GFCF_t$ represent quarter-over-quarter log differences. ΔIR_t is the first difference of the quarterly interest rate level. π_t represents annual inflation, calculated as the year-over-year percentage change of the CPI: $100 \times \left(\frac{CPI_t - CPI_{t-4}}{CPI_{t-4}}\right)$.

Table 1: Model Specifications and Sample Periods

Model	**Exogenous Vector ($\mathbf{X}_t$)**	**Sample Period**
1a	$[TBV_t]$	Q2 1961 – Q3 2023
1b	$[TBV_t]$	Q2 1985 – Q3 2023
2a	$[TBV_{t-1}]$	Q2 1961 – Q3 2023
2b	$[TBV_{t-1}]$	Q2 1985 – Q3 2023
3	$[TBV_t, EPU_t, OPU_t]'$	Q2 1985 – Q3 2023
4	$[TBV_{t-1}, EPU_{t-1}, OPU_{t-1}]'$	Q2 1985 – Q3 2023

These specifications differ along several key dimensions. Models 1 and 2 include only TBV as an exogenous variable, with Model 2 lagging TBV by one period. This specification tests for potential inertia effects in how uncertainty impacts macroeconomic variables, while also mitigating potential simultaneity bias from reverse causality. Models 3 and 4 extend this framework by incorporating EPU and OPU alongside TBV, with Model 3 using contemporaneous exogenous variables and Model 4 lagging all exogenous variables by one period. Bayesian Information Criterion lag selection tests identified lag=1 as optimal for all specifications.

4 Results

4.1 Real GDP Effects

Tables 3-6 in the Appendix present the estimation results for the six VAR-X specifications. TBV demonstrates a consistent contractionary impact on GDP across all specifications. For the full sample period (1961-2023), a one standard deviation increase in TBV reduces real GDP from 0.176 percent (Model 1a) to 0.190 percent (Model 2a), equivalent to approximately 4.0-4.3 billion CAD.[3] [4]The larger coefficient in Model 2a—where TBV enters with a one-quarter lag—suggests delayed transmission, consistent with firms deferring investment under uncertainty (real options channel). Model 2a's superior explanatory power (R^2=0.144 vs. 0.083) supports this interpretation. The restricted sample (1985-2023) yields smaller magnitudes, with GDP declining 0.07-0.10 percent across Models 1b through 4. This temporal asymmetry likely reflects Canada's 1991 adoption of inflation targeting, which reduced average TBV volatility by 36 percent and dampened agent responses to uncertainty shocks. Under a credible monetary policy regime, agents may face lower baseline uncertainty and

[3]Economic magnitude estimates are based on the 2023 average Real GDP (approx. CAD 2.25 trillion) and Total Real GFCF (approx. CAD 520 billion). Data sourced from Statistics Canada Table 36-10-0434-01 and Table 36-10-0108-01, respectively.

[4]The economic magnitude of the impact is calculated by multiplying the estimated regression coefficient ($\hat{\beta}$) by the standard deviation of the independent variable (SD_{TBV}). For example, in Model 1a, the coefficient of -0.1528 is multiplied by the standard deviation of 0.0115 (from Table 2), resulting in an approximate impact of -0.00176, or -0.176%.

respond less dramatically to uncertainty shocks. Among restricted-period specifications, Model 3 (contemporaneous exogenous variables) dominates in explanatory power (R^2=0.280), suggesting that once broader uncertainty measures (EPU, OPU) are controlled for, lagging uncertainty proxies offers no additional explanatory gain for GDP.

4.2 Fixed Capital Formation

GFCF exhibits larger sensitivity to uncertainty shocks than GDP, consistent with investment's greater cyclical volatility. Over the full sample (1961-2023), a one standard deviation TBV increase reduces GFCF by 0.289-0.306 percent (Models 1a-2a), approximately 1.5-1.6 billion CAD. Both specifications yield nearly identical explanatory power (R^2=0.06), suggesting contemporaneous and lagged uncertainty have comparable predictive capacity. The restricted sample reveals divergent patterns. Model 1b shows no significant contemporaneous effect, but lagged specifications (Models 2b and 4) indicate declines of 0.51-0.68 percent. Notably, Model 4 (where TBV, EPU, and OPU all enter lagged) achieves the highest explanatory power (R^2=0.321) compared to Model 3's contemporaneous specification (R^2=0.265). This contrasts with the full sample, where lagging TBV alone offered no explanatory gain. The differential performance suggests uncertainty's transmission operates with delay, particularly when multiple uncertainty dimensions are jointly considered. When broader economic and policy uncertainty are elevated alongside monetary uncertainty, firms' wait-and-see behavior may extend across multiple quarters. Notably, for Models 3 and 4, both EPU and OPU negatively affect GFCF, reinforcing Canada's sensitivity to broader policy uncertainty and commodity market volatility given its oil export dependence.

4.3 Nominal Variables

The relationship between TBV and both inflation and short-term interest rates proves less definitive. For interest rates, only Models 2a, 1b, and 3 exhibit statistically significant effects, but coefficients carry inconsistent signs across specifications. This ambiguity likely reflects endogeneity, considering that uncertainty may prompt central bank policy responses, but elevated uncertainty also arises from unexpected policy shifts. This creates identification challenges that reduced-form VAR-X specifications cannot fully resolve. For inflation, I find no statistically significant TBV coefficients across any specification. This null result may reflect offsetting demand-side (deflationary) and supply-side (inflationary) mechanisms, or inflation's high autocorrelation (R^2=0.80-0.95) leaving limited residual variation for uncertainty to explain. Disentangling these competing mechanisms requires structural identification beyond this paper's scope and presents an avenue for future research.

5 Conclusion

This paper investigates interest rate uncertainty's impact on the Canadian economy using quarterly volatility in 3-month treasury bill returns. Estimating six VAR-X specifications over both the full historical sample (1961-2023) and a restricted period (1985-2023), I find that heightened uncertainty exerts a consistent contractionary effect on GDP (0.07-0.19 percent) and GFCF (0.29-0.68 percent), with investment exhibiting greater sensitivity. In addition, when multiple uncertainty dimensions are considered jointly, lagged specifications outperform contemporaneous ones for investment, supporting real options theory whereby firms extend wait-and-see behavior when facing elevated uncertainty. However, while the GDP contractionary effects are larger over the full historical sample—consistent with inflation targeting reducing sensitivity after 1991—investment exhibits the opposite pattern with larger effects in the restricted period, suggesting that different transmission mechanisms warrant further investigation. In addition, the relationship between uncertainty and nominal variables remains unresolved. Interest rates exhibit inconsistent signs, likely reflecting bidirectional causality with policy responses, while inflation shows null effects—likely from offsetting demand/supply channels or high autocorrelation limiting residual variation. Future research should employ structural identification methods to disentangle these bidirectional relationships and investigate transmission channels specific to Canada—particularly whether uncertainty operates primarily through precautionary savings, financial frictions, or investment deferrals—and examine why GDP and investment exhibit divergent temporal sensitivity patterns. Understanding these mechanisms would inform whether central banks should respond to uncertainty shocks directly or focus on anchoring expectations to reduce baseline uncertainty.

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Appendix

A. TBV Calculation Methodology

To calculate Treasury Bill Volatility (TBV), I follow a three-step procedure. First, I compute the daily price of a 90-day treasury bill:

$$P_t = 100 \times \left(1 - \frac{d_t \times r}{360}\right) \tag{2}$$

where r is time-to-maturity and d_t is the discount yield. Second, I calculate the day-over-day holding period return:

$$HPR_t = 100 \times \frac{P_t - P_{t-1}}{P_{t-1}} \tag{3}$$

Finally, TBV is calculated as the standard deviation of daily holding period returns within each quarter:

$$TBV_t = \sqrt{\frac{1}{N_t - 1} \sum_{d=1}^{N_t} (HPR_{t,d} - \overline{HPR_t})^2} \tag{4}$$

where t indexes quarters, d indexes trading days within quarter t, N_t is the number of trading days in quarter t (typically 60-65 days), $HPR_{t,d}$ is the daily holding period return on day d of quarter t, and $\overline{HPR_t}$ is the average daily return within quarter t.

B. Summary Statistics and Regression Results

Table 2: Summary Statistics and Data Sources

Variable	**N**	**Mean**	**SD**	**Min**	**Max**	**Source**
GDP (10^6)	251	978,935	451,761	293,535	1,891,478	Statistics Canada
GFCF (2017 $)	251	277,345	150,084	65,003	536,777	Statistics Canada
π (%)	251	3.79	3.02	-0.86	12.70	Statistics Canada
Interest Rate (%)	251	5.74	4.00	0.18	20.74	FRED (St. Louis Fed)
TBV (%)	251	0.0121	0.0115	0.001	0.074	Bank of Canada
EPU Index	468	154	104	29	678	Baker et al. (2016)
OPU Index	420	99	82	4	500	Abiad & Qureshi (2023)

Note: All variables are at quarterly frequency. GDP, GFCF, π, and TBV were obtained at quarterly frequency. EPU and OPU were aggregated from monthly data by calculating the 3-month average.

Table 3: VAR-X Results: Real GDP (Δ GDP)

Regressor	**Model 1a**	**Model 2a**	**Model 1b**	**Model 2b**	**Model 3**	**Model 4**
Δ GDP$_{t-1}$	-0.301***	-0.306***	-0.150*	-0.160*	-0.214***	-0.205**
	[0.073]	[0.073]	[0.082]	[0.083]	[0.080]	[0.086]
Δ GFCF$_{t-1}$	0.068*	0.062*	0.096***	0.094***	0.086***	0.085***
	[0.036]	[0.036]	[0.019]	[0.019]	[0.018]	[0.020]
Δ IR$_{t-1}$	-0.042	-0.023	-0.048	-0.025	-0.049	-0.021
	[0.086]	[0.086]	[0.058]	[0.058]	[0.055]	[0.058]
π_{t-1}	0.002	0.002	-0.018	-0.019	-0.031	-0.032
	[0.027]	[0.027]	[0.027]	[0.028]	[0.027]	[0.029]
TBV$_t$	-0.153**		-0.069**		-0.092***	
	[0.070]		[0.035]		[0.034]	
TBV$_{t-1}$		-0.165**		-0.064*		-0.076**
		[0.070]		[0.035]		[0.036]
EPU					-1.78e-05***	-1.07e-05*
					[4.78e-06]	[5.50e-06]
OPU					-7.42e-06	-2.08e-07
					[5.24e-06]	[5.64e-06]
Constant	0.011***	0.011***	0.008***	0.008***	0.012***	0.010***
	[0.001]	[0.001]	[0.001]	[0.001]	[0.001]	[0.002]
R^2	0.083	0.144	0.192	0.188	0.280	0.211

$^*p < 0.10,\ ^{**}p < 0.05,\ ^{***}p < 0.01$

Table 4: VAR-X Results: Fixed Capital Formation (Δ GFCF)

Regressor	**Model 1a**	**Model 2a**	**Model 1b**	**Model 2b**	**Model 3**	**Model 4**
Δ GDP$_{t-1}$	-0.323**	-0.331**	0.770**	0.648*	0.557	0.303
	[0.151]	[0.151]	[0.355]	[0.348]	[0.350]	[0.346]
Δ GFCF$_{t-1}$	0.237***	0.227***	0.336***	0.322***	0.292***	0.228***
	[0.074]	[0.074]	[0.081]	[0.079]	[0.079]	[0.079]
Δ IR$_{t-1}$	-0.006	0.026	-0.133	-0.037	-0.136	-0.013
	[0.177]	[0.176]	[0.251]	[0.243]	[0.241]	[0.231]
π_{t-1}	-0.034	-0.034	-0.056	-0.038	-0.084	-0.038
	[0.055]	[0.055]	[0.119]	[0.116]	[0.118]	[0.115]
TBV$_t$	-0.252*		-0.154		-0.257*	
	[0.144]		[0.151]		[0.148]	
TBV$_{t-1}$		-0.266*		-0.444***		-0.595***
		[0.144]		[0.147]		[0.144]
EPU					-6.18e-05***	-5.51e-05**
					[2.10e-05]	[2.22e-05]
OPU					-4.84e-05**	-7.45e-05***
					[2.30e-05]	[2.27e-05]
Constant	0.013***	0.013***	0.003	0.006	0.019***	0.026***
	[0.003]	[0.001]	[0.004]	[0.004]	[0.006]	[0.007]
R^2	0.060	0.062	0.192	0.237	0.265	0.321

$^*p < 0.10,\ ^{**}p < 0.05,\ ^{***}p < 0.01$

Table 5: VAR-X Results: Interest Rates (Δ IR)

Regressor	Model 1a	Model 2a	Model 1b	Model 2b	Model 3	Model 4
Δ GDP$_{t-1}$	0.011	-0.000	0.532***	0.515***	0.567***	0.505***
	[0.052]	[0.052]	[0.106]	[0.109]	[0.109]	[0.114]
Δ GFCF$_{t-1}$	0.089***	0.086***	0.054**	0.052**	0.058**	0.048*
	[0.026]	[0.025]	[0.024]	[0.025]	[0.025]	[0.026]
Δ IR$_{t-1}$	0.231***	0.231***	0.120	0.105	0.121	0.104
	[0.061]	[0.060]	[0.075]	[0.076]	[0.075]	[0.077]
π_{t-1}	-0.000	0.020	-0.068*	-0.061*	-0.058	-0.054
	[0.019]	[0.019]	[0.036]	[0.036]	[0.037]	[0.038]
TBV$_t$	0.055		0.087*		0.096**	
	[0.049]		[0.045]		[0.046]	
TBV$_{t-1}$		-0.098*		-0.013		-0.021
		[0.049]		[0.046]		[0.048]
EPU					9.52e-06	3.87e-07
					[6.54e-06]	[7.32e-06]
OPU					2.72e-07	-6.90e-06
					[7.17e-06]	[7.51e-06]
Constant	-0.001	-0.000	-0.004***	-0.003*	-0.005***	0.002
	[0.001]	[0.001]	[0.001]	[0.000]	[0.002]	[0.002]
R^2	0.134	0.144	0.269	0.249	0.281	0.254

$^*p < 0.10,\ ^{**}p < 0.05,\ ^{***}p < 0.01$

Table 6: VAR-X Results: Inflation (π_t)

Regressor	Model 1a	Model 2a	Model 1b	Model 2b	Model 3	Model 4
Δ GDP$_{t-1}$	0.041	0.042	0.216*	0.221*	0.188	0.202
	[0.044]	[0.044]	[0.117]	[0.118]	[0.120]	[0.123]
Δ GFCF$_{t-1}$	0.034	0.035	0.019	0.020	0.021	0.017
	[0.022]	[0.022]	[0.027]	[0.027]	[0.027]	[0.028]
Δ IR$_{t-1}$	0.132**	0.129**	0.062	0.061	0.061	0.063
	[0.051]	[0.051]	[0.083]	[0.082]	[0.082]	[0.083]
π_{t-1}	0.964***	0.965***	0.887***	0.885***	0.872***	0.874***
	[0.016]	[0.016]	[0.039]	[0.039]	[0.040]	[0.041]
TBV$_t$	0.026		-0.007		-0.003	
	[0.042]		[0.050]		[0.050]	
TBV$_{t-1}$		0.023		0.014		0.012
		[0.042]		[0.050]		[0.052]
EPU					-6.60e-06	-6.17e-06
					[7.16e-06]	[7.91e-06]
OPU					9.70e-06	3.95e-06
					[7.85e-06]	[8.12e-06]
Constant	0.001	0.001	0.001	0.001	0.012	0.002
	[0.001]	[0.001]	[0.001]	[0.001]	[0.002]	[0.002]
R^2	0.946	0.946	0.799	0.799	0.802	0.800

$^*p < 0.10,\ ^{**}p < 0.05,\ ^{***}p < 0.01$